



OFFICIAL

Induction of Labour Guideline

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1. Purpose

The purpose of this guideline is to guide the IOL process and provide assistance in making decisions about appropriate and timely intervention, alongside the wishes and preference of each woman.

Induction of labour (IOL) is the initiation of labour in a pregnant woman who is not in labour. IOL may be indicated when the available evidence suggests benefits of birth to the mother and/or fetus outweigh the potential risks of continuing the pregnancy.

2. Indications for induction of labour

Table 2: Indications for IOL

Pregnancy	
Pregnancy beyond 41 weeks	No form of increased antenatal monitoring has shown to reduce perinatal mortality associated with prolonged pregnancy. Following 41/40 fetal wellbeing assessment using CTG and USS (umbilical artery PI and Amniotic fluid parameters are reported on) Post term induction should be offered 41+3. Exact timing depends on specific risks, individual preference, and local circumstances. Pregnancy exceeding 42 weeks to be referred to Specialist Obstetrician. .
Prelabour rupture of membranes at term	If a woman has prelabour rupture of membranes (PROM) ≥ 37 weeks gestation, and is GBS positive, commence intravenous antibiotics and if no labour has commenced after four hours of PROM, induce with a formal ARM and intravenous Syntocinon infusion. If the woman is GBS negative and both the woman and fetus are well, expectant management is offered with a time to return for induction with Syntocinon infusion at 18 to 24 hours and commencement of prophylactic intravenous antibiotic therapy for PROM. Consider PGE2 if cervix unfavourable. Consider oral Misoprostol 25 micrograms if no other risk factor Refer to AKG Pre-Labour Rupture of Membranes > 37 Weeks (PROM) Guideline
Preterm prelabour rupture of membranes	If a woman has preterm prelabour rupture of membranes before 35 weeks (PPROM), the obstetric team should discuss the following factors with her before a decision is made about whether to induce labour: • Discuss maternal and fetal risks with the woman e.g. sepsis, possible need for caesarean section problems relating to preterm birth • Local availability of neonatal intensive care facilities. Refer to AKG Premature Pre-Labour Rupture of Membranes (PPROM) Guideline
Previous Caesarean section	• The use of the Foley's /cooks balloon catheter is preferred for women requiring cervical ripening. • ARM and syntocinon as per on call specialist.

	- Expectant management is on an individual basis, with consideration to the woman's circumstances and wishes. - Women should be informed of the following risks with induction of labour: - Increased risk of need for emergency caesarean section during induced labour. - Increased risk of uterine rupture.
Fetal Death in Utero (FDIU)	Refer to AHS: Perinatal Loss Guideline
Maternal	
Ethnicity	There is currently insufficient evidence for IOL on ethnicity alone, but is currently under investigation. Consider a woman's ethnicity in the context of other risk factors when determining timing of IOL.
Maternal request	Consider IOL at term based on exceptional circumstances of the woman and her family (i.e., not solely because of patient or health care provider preference). This is not an indication for IOL
Hypertensive disorders	It is not recommended to offer birth before 37 weeks to women with mild to moderate gestational hypertension with or without treatment. Discuss with specialist obstetrician about plan of management if delivery not contemplated. For women with well controlled, stable, uncomplicated hypertension, initiate shared decision-making regarding birth from 39+0 weeks to avoid worsening of disease. For women with PET. Discuss risks and benefits of delivery with patient if preterm and delivery is required due to severity of hypertension. Earlier birth should be considered on an individualised basis in the context of suboptimal control of blood pressure or other complications. In asymptomatic well women with good blood pressure control and no features of severe disease, initiate shared decision-making regarding birth from 38+0 weeks. Refer to AKG Hypertensive Disorders of Pregnancy Guideline.
Gestational Diabetes	For well controlled GDM IOL is offered at 40-41 weeks. For women with suboptimal glycaemic control with no other risk factors IOL timing with consultant on call. Preferably 39/40. Management for women with suboptimal glycaemic control or other complications needs to be individualised, with the aim to achieve 38+0 weeks where safe to do so. Discuss possibility of ELUSC if EFW is 4500g+. Gestational Diabetes Mellitus and Gestational Diabetes Mellitus Requiring Pharmacological Therapy- AKG Guideline
Cholestasis	Between 19- <40 bile salts consider IOL at 40 weeks

	Between $\geq 49-99$ bile salts consider IOL around 38-39 weeks Bile salts >100 , IOL at 35-36 weeks
Advanced maternal age	For women aged 40 years or older, offer IOL at between 39-40 weeks gestation, CTG from 38 weeks along with ultrasound surveillance
IVF pregnancy	Offer IOL at between 39-40 weeks gestation
Fetal	
Decreased fetal movements (DFM)	Women who present on two or more occasions with DFM may have an increased risk of poor perinatal outcome, compared with those who only present once. There are no studies to determine whether intervention (e.g., delivery) alters perinatal morbidity or mortality in women presenting with recurrent DFM. Therefore, the decision whether or not to induce labour at term in a woman who presents recurrently with DFM when the growth, liquor volume and CTG appear normal must be made after careful specialist led counselling of the pros and cons of induction on an individualised basis. While the adverse outcomes of preterm birth are well understood, it is becoming increasingly apparent that early term birth (37-38 weeks' gestation) is also associated with increased short- and longer-term mortality and morbidity and poorer developmental outcomes Refer to AKG Management of Decreased Fetal Movements Guideline .
Suspected fetal compromise	Individualised specialist plan.
Fetal growth restriction/ SGA	Severity affects the decision concerning mode and timing of birth. Use of umbilical artery, middle cerebral and ductus venous doppler may assist in improving perinatal outcome through more appropriate timing of birth.
Fetal macrosomia	IOL if fetal macrosomia. If estimated fetal weight, or AC $>97\%$. Estimated fetal weight to be plotted on customised growth chart and AC to be checked on population charts. Consider IOL for estimated fetal weight or AC $>90\%$ if associated risk factors e.g GDM, previous 3 rd degree tear. IOL to be done at 40/40 NOT before.
Twin pregnancy (uncomplicated)	Monochorionic MCMA twin pregnancy elective birth from 36w+0 with antenatal corticosteroids. Dichorionic DCDA uncomplicated pregnancies offer elective birth at 37-38w.

2.1 Contraindications to IOL

Contraindications to IOL are consistent with those for vaginal birth, including:

- Placenta praevia
- Vasa praevia
- Malpresentation including transverse lie, compound presentation, and oblique lie
- Known Cephalopelvic disproportion
- Acute fetal compromise
- Maternal refusal
- Cord presentation
- Active genital herpes – management based on individual clinical cases
- Cervical carcinoma
- Footling breech
- Classical caesarean scar
- Abnormal CTG. This is not an absolute contraindication.

3. Consent

Healthcare professionals should discuss and document the following points with women being offered induction of labour:

- The reasons for induction being offered.
- Maternal and/or fetal benefit and risk.
- Proposed method/s of IOL
- When, where and how induction could be carried out.
- Recommended fetal monitoring.
- That there may be a delay to start IOL due to various reasons e.g., availability of LW beds, availability of staff and acuity
- Options for support and pain relief (recognising that women are likely to find induced labour more painful than spontaneous labour).
- Options if the woman declines IOL. It is important to respect the woman's decision and plan ongoing monitoring.
- Options if the IOL is unsuccessful
- If the discussion is at the 40-week antenatal visit, the management of prolonged pregnancy should be discussed with the woman with the recommendation that current evidence suggests pregnancy should go no longer than 42 weeks gestation.

The woman's informed consent for IOL must be acquired at the clinic/AAU appointment in which the IOL is booked. Written consent should be gained by a doctor/midwife and must be documented on the 'Patient consent to treatment or investigation – adult or mature minor' AKMR30 form and filed in the woman's hospital based medical records either at time of booking, or on admission. Documentation of date and time should also be written in their hand-held record and/or AAU MR8

Women should be directed to the Pregnancy, Birth and your Baby book given to them at antenatal booking for written information around IOL pages 44-47 inclusive.

3.1 Woman declines/refuses IOL

If following informed discussion and recommendation a woman declines IOL, a non –standard management plan sticker must be completed and documented in the medical record by a Specialist obstetrician/GPO. An individualised plan of care to be formulated and offer a CTG to assess baby's wellbeing.

4. Process for booking IOL

The practitioner who discusses and recommends IOL to a woman is responsible for contacting the Clinical Midwife (CM) co-ordinator of Antenatal Assessment Unit (AAU). This should be in consultation with the woman, ideally when she is present in the clinics or AAU. The CM co-ordinator in AAU will access the appropriate scheduling of IOL in the IOL book and schedule the IOL on a mutually agreeable date/time.

Noting that a maximum of **three** inductions of labour are to be booked, should the requested date/time not be available the CM co-ordinator and requesting practitioner will book the next most appropriate date/time for IOL. If there is no suitable date/time available the request should be escalated to the Associate Midwifery Manager – Non-Ambulatory Services and/or HOD Obstetrics and Gynaecology in hours or Specialist Obstetrician on call out of hours to resolve or escalate, as required. Any IOL before 39 weeks needs to be discussed with Midwifery Manager (MM), and GP Obstetrician of the day. The reason and agreement need to be documented on AKMR 5.1 form. (Appendix1). This is part of the Preterm Birth Alliance initiative to reduce early term births by 20%

It is the responsibility of the requesting practitioner to communicate the booked date/time to the woman and ensure she knows where and when to present. Documentation of the date/time of the IOL should be made in the woman's National Handheld Pregnancy record.

- If the level of activity on the maternity unit is high, it may be necessary to prioritise IOL according to clinical need which may mean delaying booked induction.
- There will need to be a collective discussion in the AAU with the maternity team involving Midwifery Manager, Midwives, GP Obstetrician and Specialist Obstetrician on call around the acuity and bed availability
- Prioritisation of IOL will be by traffic light system of green, amber and red. If there are two reds, then there will be collective discussion with on call team and two red dots can be inserted against the IOL most urgent (IOL AKMR 5.6 form) (Appendix 2)

4.1 Prioritisation of IOL

Table 1: IOL traffic light prioritisation

RED	AMBER	GREEN
- Confirmed Intrauterine - Growth Restriction (IUGR) - Pre-Eclampsia - Suspected Fetal Compromise - Fetal Demise	- Gestational Diabetes - Post term women with previous caesarean section (NBAC) - Prolonged Rupture of Membranes - Confirmed Macrosomia EFW or AC>97%. - AC or EFW>90% with other risk factors	- Uncomplicated Post Term Pregnancies between 41-42 weeks gestation - Induction for social reasons

4.2 Management of Delayed Booked Induction

- If the maternity unit is at capacity and IOL is not feasible, the maternity team will provide the woman with an explanation, she should be offered an explanation and a full antenatal examination including a CTG and possibly an USS depending on circumstances.
- If both the woman and fetus are well, she may go home to be called when the unit is able to care for her
- The woman will be offered a full antenatal examination and CTG daily until labour can be induced
- No woman should be delayed if the induction is for urgent medical reasons purely because of staff/ ward acuity. This decision should be escalated to Midwifery Managers to acquire extra staffing and/ or transfer to another centre if not possible.

5. Admission for IOL

If a pregnancy is deemed as very high risk, induction should take place in birth suite where 1:1 care can be facilitated. All other IOL should be commenced in the Antenatal Assessment Unit (AAU) and remain there until the woman is in active labour or requires intravenous Syntocinon therapy, and/or continuous CTG monitoring. This encourages the woman to mobilise.

5.1 Pre IOL assessment

- Review the maternal history and confirm gestation
- Perform baseline maternal observations (T, P, BP, RR and O2 sats) and document these on MADDs chart.
- Perform abdominal palpation and confirm presentation, lie, position, and engagement
- Assess the cervix. The findings are circled and given a Bishop Score (Table 1)
- Confirm membrane status (ruptured or intact)
- Assess fetal wellbeing: Confirm normal CTG. If abnormal, escalate as per EMHS Fetal Surveillance and CTG monitoring Policy.

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- Assess for contraindications to IOL
- Consider urgency for IOL
- Insert a 16g Peripheral Intravenous Cannula and document on the PIVC AKMR 121.6 form
- Send bloods for FBC and serum hold. Consider cross match if further risks identified.

5.2 Cervical assessment.

The bishop score is commonly used to assess the cervix and to inform the choice of method of induction. Each feature of the cervix is scored and then the scores are summed. The state of the cervix is one of the important predictors of successful IOL. The cervix is deemed unfavourable if the bishops' score is ≤ 6 .

Table 3: Bishop Score

Calculate Modified Bishop Score			
Bishop score	0	1	2
Station	-3cm	-2cm	-1cm
Dilation	0cm	1-2cm	3-4cm
Length	3cm	2cm	1cm
Consistency	Firm	Medium	Soft
Position	Posterior	Mid-position	Anterior

5.3 Methods of cervical ripening and IOL

5.3.1 Sweeping membranes for reducing formal IOL rate.

At the 38-week antenatal visit, all women should be offered information about the risks associated with pregnancies that last longer than 42 weeks, and their options. There is no evidence to support repetitive membrane sweeping from 38 weeks in an otherwise normal pregnancy. The optimal gestation to commence is controversial; however, evidence suggests 40-41 weeks in nulliparous and 41 weeks for parous women. Membrane sweeping involves the examining finger passing through the cervix to rotate the wall of the uterus to separate the chorionic membrane from the decidua.

If the cervix will not admit a finger massaging around the cervix in the vaginal fornices may achieve a similar effect. Information should include membrane sweeping at 41 weeks gestation makes spontaneous labour more likely and may reduce the need for formal induction of labour. It should also be highlighted that discomfort, prelabour spontaneous rupture of membranes, vaginal bleeding and a 'show' are possible from the procedure. Prior to offering this procedure obtain the woman's consent.

5.4 Breast stimulation

There is insufficient evidence on the efficacy of nipple stimulation and the effectiveness in ripening the cervix to assist favourability for IOL. Women can use nipple stimulation to be proactive in their induction process. Stimulation of the nipple's releases oxytocin, which in turn

contracts the uterus. Used in conjunction with a membrane sweep the woman may be proactive in assisting herself into possible labour.

6. Induction methods

Table 4:

Bishop Score 0-4	Bishop Score 5-6	Bishops Score >7
- Primiparous: 2mg Prostaglandin E2 Gel or 10mg Cervidil - Multiparous: 1-2mg Prostaglandin E2 Gel or 10mg Cervidil - Oral Misoprostol 25 micrograms. Oral Misoprostol as an induction agent should be given as per the Aarhus Protocol in below table	1mg Prostaglandin E2 Gel or 4mg Cervidil PGE2 pessary in both primiparous and multiparous women. - Transcervical balloon Catheter - Oral misoprostol 25 micrograms.	- Artificial rupture of membranes (ARM) - Oxytocin infusion

See next page for Dosage and Monitoring of Oral Misoprostol

Dosage and monitoring of Oral Misoprostol

as per WNHS guideline [Misoprostol protocol for induction of labour from 37+0 weeks](#)

- Oral misoprostol as an induction agent should be given as per the Aarhus Protocol which is summarised below:

	Inpatient regimen
Course 1 (Six doses of 25 microg misoprostol tablets taken orally two hours apart)	Clinical examination* • Misoprostol 25 microg every 2 hours, orally, for 3 doses CTG • Misoprostol 25 microg every 2 hours, orally, for 3 doses CTG after 6th dose 12-hour rest
Course 2 (Eight doses of 25 microg misoprostol tablets taken orally two hours apart)	Clinical examination*. Misoprostol regimen should be commenced if regular contractions have not started or artificial rupture of membranes (ARM) is not possible. • Misoprostol 25 microg every 2 hours, orally, for 3 doses Clinical examination* • Misoprostol 25 microg every 2 hours, orally, for 5 doses CTG after 14th dose 10-hour rest
Course 3 (Three doses of 25 microg misoprostol tablets taken orally two hours apart)	Clinical examination*. Misoprostol regimen should be commenced if regular contractions have not started or ARM is not possible. • Misoprostol 25 microg every 2 hours, orally, for 3 doses 2-3 hours after 17th dose: Clinical examination* Discuss with Consultant Obstetrician regarding further IOL options. NOTE: Further ripening should only be commenced after a rest period of 24 hours. During this time a CTG should be performed at 12 and 24 hours unless otherwise indicated.
* ‘Clinical examination’ includes: vaginal examination, abdominal palpation and CTG	

7. Dinoprostone

Vaginal gel (prostaglandin E2 (prostin) 1mg and 2mg.

Controlled release vaginal pessary (Cervidil) 10mg

Table 5: (Quick reference guide see appendix 2)

Indication	- Unfavourable cervix - May be used following balloon catheter when there has been no/minimal effect on cervical ripening and ARM is not possible.
Contraindication	- Known hypersensitivity to PGE2 or any other constituent of the pessary >5 previous births - Concurrent oxytocin use, or intravenous oxytocic within 30 minutes - History of previous major uterine surgery - Cephalopelvic disproportion - Abnormal CTG or suspected fetal compromise - Malpresentation - Unexplained vaginal discharge and/or abnormal uterine bleeding during the current pregnancy - Where vaginal delivery is not indicated such as vasa praevia or active herpes - Use with caution: asthma or COPD, glaucoma, epilepsy, women at risk of DIC, compromised cardiovascular/ hepatic/renal/pulmonary system
Administration	- Women report better satisfaction if administered in the morning. - Pessary use may avoid repeated administration. - Gel may be more appropriate when cervix is unfavourable.
Dinoprostone Gel	- Remove from the fridge and stand for 30 minutes prior to insertion. - Use water-soluble lubricants (not obstetric cream). - Insert high into the posterior fornix. - Not for intracervical administration. - Advise recumbent or left lateral for 30 minutes post insertion.
Dinoprostone Pessary (Cervadil)	- Remove from fridge and use immediately, warming is not required. - Use water-soluble lubricant (not obstetric cream). - Insert and position transversely in the posterior fornix. - Advise recumbent or left lateral for 30 minutes post insertion. - Advise to report if it falls out.
Monitoring, Post protocol Insertion	- CTG for minimum 30 minutes. Discontinue if normal. Continue if abnormal and escalate as per protocol. - T,P,R,BP and saturation record on MADDs chart as per protocol, continue 4 hourly observations and FH - CTG prior to next assessment (6 hours for gel) 6,12,18,24, respectively for cervadil.

	When in active labour, continuous CTG.
Assessment of Progress	- Reassess BS 6 hours post insertion with gel and repeat if ARM not possible. - Reassess BS at least 12 post Cervadil insertion. If ARM not possible leave for a further 6 hours. - Cervadil to be removed after 24 hours.
Indications for Removal Cervadil	- Onset of regular painful contractions. - Spontaneous rupture of membranes. - Hyperstimulation. Maternal systemic adverse effects (nausea vomiting hypotension, tachycardia). - If starting oxytocin (remove at least 30 minutes prior to commencement).

8. Transcervical Foley's or Cook's Balloon catheter

The use of the transcervical Foley's or Cook's Balloon catheter for cervical ripening is an efficient, safe, cost effective, reversible method of induction of labour, and is associated with a low incidence of uterine contractile abnormalities. It provides a non-pharmacological method of induction of labour in situations where birth is indicated but not urgent. **Women can choose to go home with balloon in situ**

Table 6: (Quick reference guide refer to appendix 3)

Contraindications	- Ruptured membranes - Malpresentation - Placenta or vasa praevia - Simultaneous use of prostaglandins
Indications	- It provides an option for cervical ripening when there are contraindications to pharmacological agents. 27,60. - TOLAC - IUGR - Grand multipara - Oligohydramnios or when pharmacological agents have failed.
Pre procedure	- Obtain consent - Ensure normal CTG - Ensure an empty bladder - Perform abdominal palpation to confirm presentation and lie. - Assist the woman into lithotomy position
Transcervical Foley's or Cook's Balloon procedure	Using a Bi-valve Cuscoe speculum, lubricating gel and sponge forceps under good lighting, insert a Foley's or Cook's catheter into the internal os, keep the catheter secured while inflating to prevent it falling out during inflation.

	• Inflate the Foley's single balloon with 60mLs sterile water and place a spigot in the end. Inflate the Cook's double balloons with 80mLs sterile water in each. • Tape the catheter to the woman's leg for her comfort. • If the woman experiences pain or bleeding during the inflation of the balloon, stop inflation and ensure the catheter has been inserted deep enough. • Auscultate the fetal heart after the procedure and record findings in progress notes MR55A. • Assess after 12 hrs as a Cook's Balloon catheter is recommended to be removed after 12 hrs. • In a Foley's, reassess after 12 hrs and ensure the balloon is still in the correct place. The recommendation is to place traction on the catheter and tape it firmly to the woman's leg. • Remove the Foley's after deflating the balloon 18 to 24 hrs post insertion.
Monitoring	• Immediately following balloon insertion: T, P, R, Bp, sats FH, and vaginal loss. • CTG is not required unless other indications (e.g. uterine activity). • Ongoing monitoring as for latent first stage of labour.
Early removal	• Observations and/or FH are abnormal. • Persistent pain. • Remove if there is frank bleeding • Remove if there is spontaneous rupture of membranes. • Remove if labour commences – regular, painful uterine action with cervical changes
If balloon catheter falls out	• Perform VE. • Perform ARM and oxytocin ASAP (due to the temporary dilatory effect of balloon catheters).
Removal of balloon catheter	• Cooks- After 12 hours remove the balloon catheter by completely deflating the balloons. Do not leave longer than 18 hours. • Foleys-after 24 hours remove the balloon catheter by completely deflating the balloon. • Once the balloon catheter has been removed perform an ARM and commence an oxytocin infusion. • Refer to AHS labour guidelines.

9. Artificial rupture of membranes (ARM)

Table 7: (Quick reference guide refer to appendix 4)

Indication	Favourable cervix BS>7
Contraindications and associated risks	- Placenta praevia. - Active herpes lesions. - Vasa praevia/cord presentation. - Poor application of presenting part/unstable lie. - High head 5/5 palpable abdominally. - Polyhydramnios. - PGE2 gel within the last 6 hours. - Cervadil within last 30 minutes.
Procedure	- Obtain consent. - Ensure an empty bladder. - Perform and abdominal palpation to determine presentation and lie. - Auscultate fetal heart. (Ensure normal). - Perform a VE to confirm presentation and station. - Using an amnihook, slide the instrument alongside the fingers with the hook facing down, and then rotate the hook to face upwards to rupture the membranes. - Using an amnicot, apply over the middle/index finger, and face it downwards when inserting into the vagina, then turn the hook upwards to rupture the membranes. - Check for amniotic fluid after one or two well applied scratches. - Place your finger through the hole in the membranes and strip the membranes back in as wide a circle as possible ensuring the presenting part is well applied to the cervix. - Note the amount, colour, odour and consistency of the amniotic fluid.
Monitoring	- Providing the liquor is clear Auscultate the fetal heart for a full minute. - Commence continuous CTG monitoring if there is meconium, frank blood, or oxytocin has been commenced. - T, P, R, Bp and sats as per policy. - Encourage mobilisation to promote the onset of uterine contractions.

10. Oxytocin infusion for induction / augmentation of labour

Table 8: (Quick reference guide refer to appendix 5)

Indication	• IOL in the setting of ruptured membranes
Contraindications	• Intact membranes • 2 or more previous caesarean sections • Use with caution: Grande multiparity, 1 previous caesarean section
Cautions	• Do not commence within 6 hours of Dinoprostone gel administration • Do not commence within 30minutes of removal of Dinoprostone pessary
Before administration	• Confirm a normal CTG • If membranes are intact perform ARM • If SROM ensure forewaters are ruptured.
Monitoring	• Provide one to one midwifery care • Continuous fetal monitoring by CTG. Ensuring 30-minute reviews and 2 hourly fresh eyes. Escalate abnormal CTG as per policy. • Monitoring uterine contraction by palpation is essential • Commence partogram and follow prompts as per protocol escalate deviations as per policy. • Monitor fluid balance • Encourage the woman to empty her bladder every three hours. If unable to do so and the bladder is palpable, pass an in/out catheter. Depending on the woman's mobility and ability to pass urine herself, assess if an indwelling catheter (IDC) is required • If epidural is sited, insert an IDC.

10.1 Dosage of Oxytocin

A standard dilution of 10 IU Oxytocin in 500ml Compound Sodium Lactate. At this dilution, an infusion rate of 3ml/hr. equates to 1 milli unit (mU) of oxytocin per minute.

The dose and rate of intravenous Oxytocin therapy must be titrated against uterine action. The aim is to achieve a contraction frequency of 3-4:10, lasting 60 seconds using the minimum dose of oxytocin possible to establish activity sufficient to produce cervical change, while avoiding Hyperstimulation.

Table 9: Oxytocin infusion conversation chart

Start Dose	Incremental Increase Oxytocin	Dosage interval
2mU per minute (6ml/hr)	Increase the infusion rate by 4mU per minute (12ml/hr.) to a maximum dose of 20mU per minute (60ml/hr.)	No less than 30 minutes (This is the time required to achieve steady-state plasma levels)

Time after starting (Minutes)	Oxytocin dose (mU per minute)	Volume infused (mls per hour)
0	2	6
30	4	12
60	8	24
90	12	36
120	16	48
150	20	60
180	24	72
210	28	84
240	32	96
270	36	108

Administration	- Deliver the intravenous oxytocin infusion through an infusion pump and ensure the giving set has a triple lumen peripheral IV (V-set) attached. - The intravenous oxytocin infusion line is then attached to the pump line preventing bolus administration of oxytocin. - Where the oxytocin infusion is to run as a sideline to a main intravenous line, it should be connected to the main line with a V-Set. The V-Set acts as an anti-reflux valve preventing bolus administration of oxytocin - Once the maximum dose of 20mU/minute (60ml/hr.) has been reached and has been infusing for 30 minutes without sufficient/efficient uterine contractions, inform the Specialist Obstetrician to review the woman prior to higher doses being administered. The maximum dose should not exceed 36mU/minute (108mls/hour) - The GP Obstetrician on call should be consulted before increasing rate of intravenous oxytocin therapy in the presence of an abnormal CTG - Consider modifying the regime in grand multipara or women who have had a previous caesarean section
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11. Risks and benefits associated with IOL

Table 9: Risks and benefits

Failed induction	- Failed induction of labour does not necessarily indicate caesarean section. - Maternity clinicians should discuss this with the woman and provide support. - The woman's condition and the pregnancy in general should be fully reassessed, and fetal wellbeing should be assessed using CTG - The subsequent management options include: - Surgical Induction with ARM if possible. - Consider Foleys/cook's catheter if initial IOL was with prostaglandin gel or Cervidil pessary. - If mechanical induction fails, continuing IOL may involve Dinoprostone or reinsertion of another balloon catheter after 24 hours. - Caesarean section will need to be discussed with the woman and her birth partner if all methods of cervical ripening and IOL fail.
Tachysystole or hypertonus or hyperstimulation	- Continuous CTG. - Maternity staff remaining with the woman until normal activity is observed. - For Prostaglandin E2 Gel: Digitally remove as much of the gel as possible and give a saline vaginal wash out using a 20 ml syringe. Consider giving a subcutaneous tocolytic in the form of Terbutaline 0.25mg for women with uterine hyperstimulation. - For Cervidil PGE2 Pessary: Remove the pessary immediately and give consider giving a subcutaneous tocolytic in the form of Terbutaline 0.25mg for women with uterine hyperstimulation. - Intravenous oxytocin infusion tachysystole or hypertonus: Decrease infusion rate. - Intravenous oxytocin infusion with uterine hyperstimulation Cease Infusion - Consider terbutaline 250 mcg subcutaneously. - Escalate as per policy - Inform the GP Obstetrician/Specialist Obstetrician to review the woman. - Consider delivery of the baby in the presence of fetal heart rate abnormalities
Cord prolapse	To reduce the likelihood of cord prolapse, which may occur at the time of Amniotomy, the following precautions should be taken: Before induction, engagement of the presenting part should be assessed.

	• Obstetricians or midwives should palpate for umbilical cord presentation during the preliminary vaginal examination and avoid dislodging the baby's head. • Amniotomy should be performed with caution if the baby's head is high – the specialist obstetrician should be involved in the decision for this to occur. Operating theatre co-ordinator should be informed. • If cord prolapse occurs category 1 caesarean section should be immediately initiated. • Call 55 code blue caesarean.
Uterine rupture	• An uncommon event with IOL • A life-threatening event for mum and baby • If uterine rupture is suspected during induced labour, Category 1 caesarean section should be immediately initiated. • Call 55 Code blue caesarean.
PPH	• IOL with oxytocin has been associated with an increased risk of PPH. • Follow AHS guidelines for primary PPH. Refer to AHS: Postpartum Haemorrhage Guideline
Increased intervention	• Compared with spontaneous labour IOL may be associated with a higher incidence of additional intervention (electronic fetal monitoring, analgesia usage) although there appears to be no increase in instrumental births or CS in RCT's. • Compared with expectant management, no association found between IOL and increased rates of adverse perinatal outcomes of neonatal unit admissions, maternal death, or meconium stained amniotic fluid.
Benefits	• When performed for a valid indication IOL should result in a reduction in perinatal mortality and morbidity. • Limited evidence suggests women may prefer IOL to expectant management (serial antenatal monitoring) beyond 41 weeks.

12. Legislative context

The following documents are required to give effect to this policy:

- HDWA
 - [Clinical Handover Policy](#)
 - [Consent to Treatment Policy](#)
- EMHS
 - [Patient Identification and Procedure Matching EMHS: 21](#)

13. Supporting information

The following documents support the implementation of this policy:

- [Hypertensive Disorders of Pregnancy Guideline](#)
- [Management of Decreased Fetal Movement Guideline](#)

- [RANZCOG The use of Misoprostol in Obstetrics and Gynaecology](#)
- [Perinatal Loss Guideline](#)
- [Pre-Labour Rupture of Membranes > 37 Weeks \(PROM\) Guideline](#)
- [Premature Pre-Labour Rupture of Membranes \(PPROM\) Guideline](#)

14. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated by the Obstetrics and Gynaecology Department at the departmental safety and quality meeting.

Monitoring activities will be via clinical incident reported data via the Datix Clinical Incident Management System (CIMS).

15. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Medication Safety - Action 4.1 Integrating clinical governance
- Standard 5 – Comprehensive Care - Action 5.11 Clinical assessment and 5.13
- Standard 6 – Communicating for Safety - Action 6.7 Clinical handover
- Standard 8 – Recognising and Responding to acute deterioration - Action 8.6 Escalating care

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17. Glossary

The following definitions are relevant to this policy.

Term	Definition
Amniotomy	Artificial rupture of membranes (ARM) to initiate or speed up labour.

Term	Definition
Cervical ripening	A prelude to the onset of labour whereby the cervix becomes soft and compliant. This allows its shape to change from being long and closed, to being thinned out (effaced) and starting to open (dilate). It either occurs naturally or as a result of physical or pharmacological interventions.
Expectant management	Allowing labour to develop and progress under supervision without intervention, unless clinically indicated.
Favourable cervix	The cervix is said to be favourable when its characteristics suggest there is a high chance of spontaneous onset of labour, or of responding to interventions made to induce labour.
Fetal Growth Restriction	Also known as intrauterine growth restriction (IUGR). This indicates the presence of a pathophysiological process in utero that inhibits fetal growth.
Induction of labour	The process of artificially initiating labour.
Mechanical method	Non-pharmacological methods of inducing labour ARM, stretch and sweep, nipple stimulation, Foley's or cooks catheter.
Prolonged pregnancy	A pregnancy past 41 weeks gestation.
Transcervical balloon catheter	This can be introduced through the cervix and the balloon inflated.
Uterine hyperstimulation	Either uterine tachysystole or uterine hypertonus with fetal heart rate (FHR) abnormalities.
Uterine hypertonus	Contractions lasting more than 2 minutes in duration, or contractions occurring within 60 seconds of each other and without fetal heart rate abnormalities.
Uterine tachysystole	More than 5 contractions in 10 minutes without FHR abnormalities.

18. Document contact

Director Clinical Services

Enquiries relating to this guideline may be directed to:

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19. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 4.2	31 03 2026	07/09/2026	Minor review to align with the new GDMi policy document.

20. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	30 03 2026
Published date	31 03 2026

Appendix 1: Method of induction of labour flow chart

Induction of Labour

Indication

- Maternal and/or fetal benefit

Contraindications

- As for vaginal births

Communication with women

- Indication
- Maternal &/ or fetal benefit & risk
- Individual circumstances
- Proposed IOL methods
- Options for pain management
- Options if:
 - 1.IOL unsuccessful
 - 2.IOL declined
 - 3.Expected management preferred
- Time for decision – making
- Obtain informed consent
- Document above

Membrane sweep

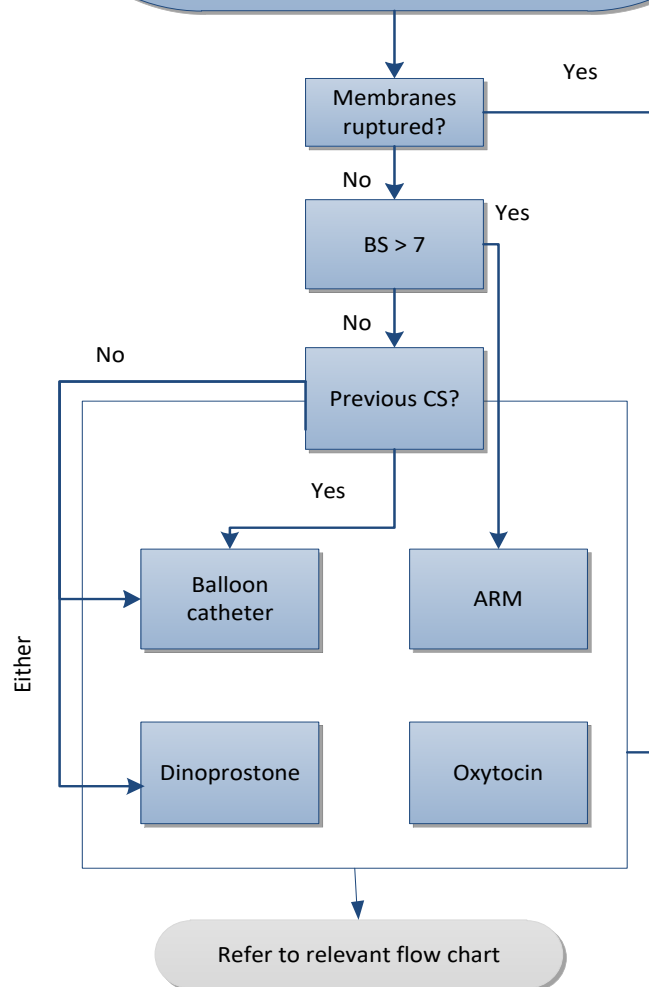
- Discuss antenatally
- Offer prior to IOL

- If IOL declined or postponed**
- Consider individual circumstances, women's preferences, local service capabilities and priorities
- Perform maternal and fetal assessment
- Arrange ongoing monitoring
- From 42 +0 weeks offer twice weekly:
 - 1.CTG
 - 2.USS for fetal wellbeing
- Provide verbal and written information on fetal movement
- Advise to contact health care provider if concerned
- Document assessment and plan of care in health records

Method

Pre IOL assessment

- Review history
- Confirm gestation
- Baseline observation
- Abdominal palpation (presentation, fundal height, position, lie, engagement)
- CTG: consult obstetrician if abnormal
- Vaginal examination:
 - 1.Assess BS
 - 2.Membrane status
- Intact
- Ruptured



Appendix 2: Prostaglandin E₂ (Dinoprostone)

Induction of Labour

Indication

- Unfavourable cervix (BS <6)
- Following balloon catheter if no/minimal effect on cervical ripening and ARM not technically possible

Contraindications

- Known hypersensitivity
- Ruptured membranes
- Multiparity > 5
- Previous CS or uterine surgery
- Malpresentation
- Undiagnosed PV bleed
- Abnormal CTG/ fetal compromise

Cautions

- Multiple pregnancy
- Asthma, chronic obstructive pulmonary disease: may cause bronchospasm
- Epilepsy
- Cardiovascular disease
- Raised intraocular pressure, glaucoma
- Avoid concurrent oxytocin use

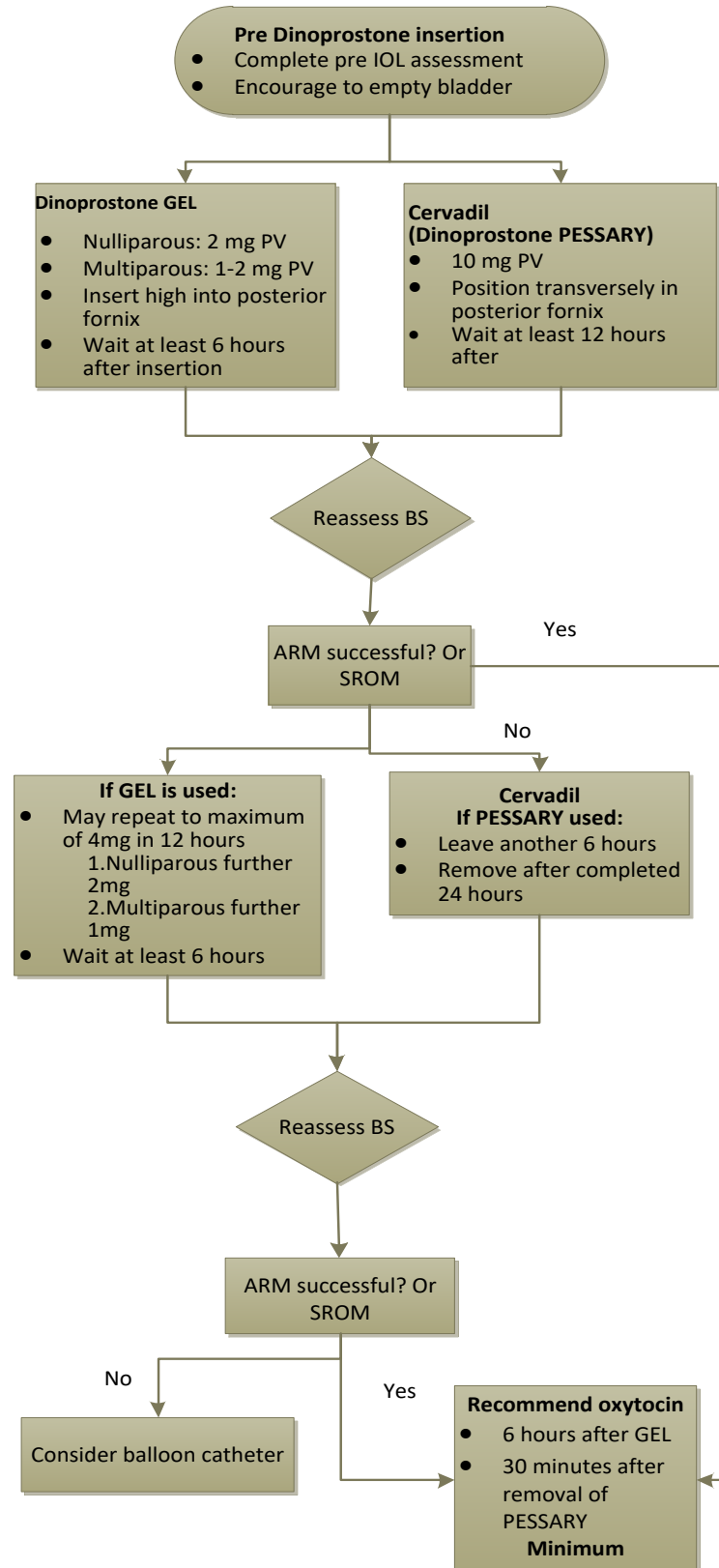
Post dose care

- TPR, BP, FHR, uterine activity, PV loss, first hour then every 4 hours
- CTG for minimum of 30 minutes
- If observations normal, no contractions and not otherwise indicated, ongoing care as for latent first stage of labour
- Continuous CTG when in active labour or when contractions are >3 in 10 minutes
- After insertion advise woman to:
 1. Remain recumbent for 30 minutes
 2. Inform staff as soon as contractions commence

PESSARY removal indications

- Onset of regular, painful uterine contractions, occurring every 3 minutes regardless of cervical change
- Ruptured membranes
- Fetal distress
- Uterine hyperstimulation or hypertonic uterine contractions
- Maternal systemic adverse effects (e.g. nausea, vomiting, hypotension, tachycardia)
- Insufficient cervical ripening after 24 hours

Method



Appendix 3: Balloon catheter flow chart

Induction of Labour vchart: Method of Induction

Inductions

- BS < 6
- Previous CS
- Following Dinoprostone if no / minimal effect on cervical ripening and ARM not technically possible
- Reduced risk of uterine hyperstimulation is desirable, e.g. SGA, IUGR, oligo

Contraindications

- Rupture membranes
- Undiagnosed bleeding
- Simultaneous use of prostaglandins
- Low lying placenta
- Polyhydramnios
- Abnormal FHM on auscultation or CTG

Relative contraindications

- Antepartum bleeding
- Lower tract genital infection
- Fetal head not engaged (4/5 or 5/5 above pelvic brim)

Balloon Catheter

Insertion procedure

- Pre catheter insertion:
 1. Insert 16G cannula
 2. Ensure pre IOL assessment complete
 3. Encourage to empty bladder
- Performed by medical or midwife staff:
 1. Contact more experienced clinician if two unsuccessful attempts
- Inflate balloon catheters with sterile water or 0.9% sodium chloride:
 1. Double balloon: 80mL each balloon
 2. Single balloon: 30 – 80 ml
- Document inflation volume

Post procedure observation and care

- Pulse, BP, FHR, uterine activity and vaginal loss
 1. Immediately and repeat at 30 minutes
 2. Medical review if malpresentation or fetal head 5/5 palpable after insertion
 3. CTG not required (unless other indications)
- If observations normal, no contraction and not otherwise indicated, ongoing care as for latent first stage of labour
- Assess for / ask about discomfort

Moderate or severe discomfort?

Labour commenced?

Yes

No

Reduce balloon volume

- Remove 10mL from each balloon
- Reassess and repeat to a minimum of 50 mL of residual volume in each balloon
- Document volume removed

No

Moderate or severe discomfort?

Yes

If not in labour,
offer analgesia and
sedation

No

Ongoing pain /
discomfort?

Yes

As required

Labour care (Birth
Suite)
Ongoing care as indicated

Any of

SROM
Catheter out?
Observations normal?
Labour commenced?

Yes

No

12 hours after insertion:
(Do not leave catheter in situ longer
than 18 hours)

- Reassess in birth suite
- Recommend ARM

ARM successful

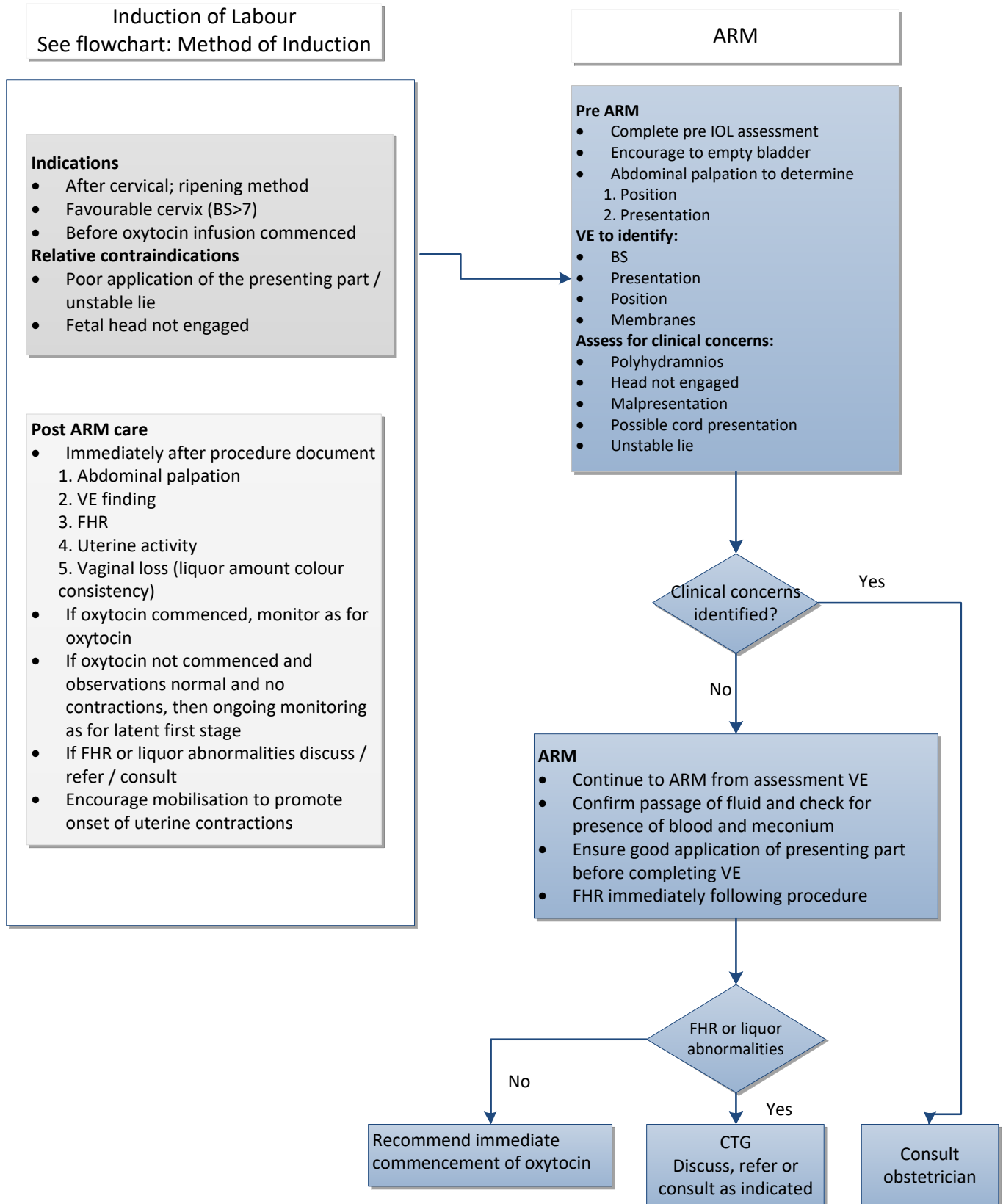
Yes

Continue IOL
Recommend
immediate
commencement of
oxytocin

No

Obstetric review
Consider:
• Dinoprostone,
or
• Reinsert
catheter after
24 hours

Appendix 4: Artificial rupture of membranes flow chart



Appendix 5: Oxytocin flow chart

